

Spring 2026 Data C100/C200 Midterm 2 Reference Sheet

Modeling

Concept	Formula	Concept	Formula
Variance, σ_x^2	$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$	Correlation r	$r = \frac{1}{n} \sum_{i=1}^n \frac{x_i - \bar{x}}{\sigma_x} \frac{y_i - \bar{y}}{\sigma_y}$
L_1 loss	$L_1(y, \hat{y}) = y - \hat{y} $	Linear regression estimate of y	$\hat{y} = \theta_0 + \theta_1 x$
L_2 loss	$L_2(y, \hat{y}) = (y - \hat{y})^2$	Least squares linear regression	$\hat{\theta}_0 = \bar{y} - \hat{\theta}_1 \bar{x} \quad \hat{\theta}_1 = r \frac{\sigma_y}{\sigma_x}$
Empirical risk with loss L		$R(\theta) = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{y}_i)$	

Multiple Linear Regression Formulas

Concept	Formula	Concept	Formula
Mean squared error	$R(\theta) = \frac{1}{n} \ \mathbb{Y} - \mathbb{X}\theta\ _2^2$	Normal equation	$\mathbb{X}^T \mathbb{X} \hat{\theta} = \mathbb{X}^T \mathbb{Y}$
Least squares estimate, if $\mathbb{X}$ is full rank	$\hat{\theta} = (\mathbb{X}^T \mathbb{X})^{-1} \mathbb{X}^T \mathbb{Y}$	Multiple R^2 (coefficient of determination)	$R^2 = \frac{\text{variance of fitted values}}{\text{variance of } y}$
Ridge Regression L2 Regularization	$\frac{1}{n} \ \mathbb{Y} - \mathbb{X}\theta\ _2^2 + \lambda \ \theta\ _2^2$	Squared L2 Norm of $\theta \in \mathbb{R}^d$	$\ \theta\ _2^2 = \sum_{j=1}^d \theta_j^2$
Ridge regression estimate (closed form)	$\hat{\theta}_{\text{ridge}} = (\mathbb{X}^T \mathbb{X} + n\lambda I)^{-1} \mathbb{X}^T \mathbb{Y}$	L1 Norm of $\theta \in \mathbb{R}^d$	$\ \theta\ _1 = \sum_{j=1}^d \theta_j $
LASSO Regression L1 Regularization	$\frac{1}{n} \ \mathbb{Y} - \mathbb{X}\theta\ _2^2 + \lambda \ \theta\ _1$		

Scikit-Learn

Package: <code>sklearn.linear_model</code>			
Linear Regression	Logistic Regression	Function(s)	Description
✓	-	<code>LinearRegression(fit_intercept=True)</code>	Returns an ordinary least squares Linear Regression model.
-	✓	<code>LogisticRegression(fit_intercept=True, penalty='l2', C=1.0)</code>	Returns an ordinary least squares Linear Regression model. Hyperparameter C is inverse of regularization parameter, C = 1/λ.
✓	-	<code>LassoCV()</code> , <code>RidgeCV()</code>	Returns a Lasso (L1 Regularization) or Ridge (L2 regularization) linear model, respectively, and picks the best model by cross validation.
✓	✓	<code>model.fit(X, y)</code>	Fits the scikit-learn <code>model</code> to the provided <code>X</code> and <code>y</code> .
✓	✓	<code>model.predict(X)</code>	Returns predictions for the <code>X</code> passed in according to the fitted <code>model</code> .
✓	✓	<code>model.predict_proba(X)</code>	Returns predicted probabilities for <code>X</code> according to the fitted <code>model</code> . If binary classes, will return probabilities for both class 0 and 1.
✓	✓	<code>model.coef_</code>	Estimated coefficients for the linear model, excluding the intercept.
✓	✓	<code>model.intercept_</code>	Bias/intercept term of the linear model. Set to 0.0 if <code>fit_intercept=False</code> .

Package: `sklearn.model_selection`

Function	Description
<code>train_test_split(*arrays, test_size=0.2)</code>	Returns two random subsets of each array passed in, with 0.8 of the array in the first subset and 0.2 in the second subset.

Gradient Descent Update Rule

Gradient Descent

For a learning rate α , the gradient update step is:

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \nabla_{\theta} L(\theta^{(t)})$$

where $\nabla_{\theta} L(\theta^{(t)})$ is the partial derivative/gradient of L with respect to θ , evaluated at $\theta^{(t)}$.